

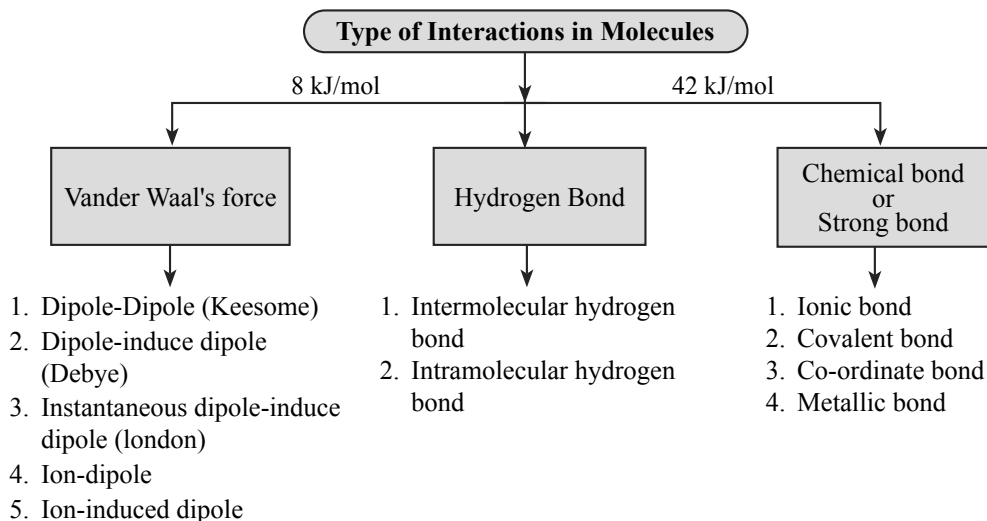
# Chemical Bonding and Molecular Structure

## SHORT NOTES

### Chemical Bond

The force of attraction which hold together atom, molecule or ions with in chemical species.

It is always exothermic process.



### Properties of Ionic Compound

#### General Properties

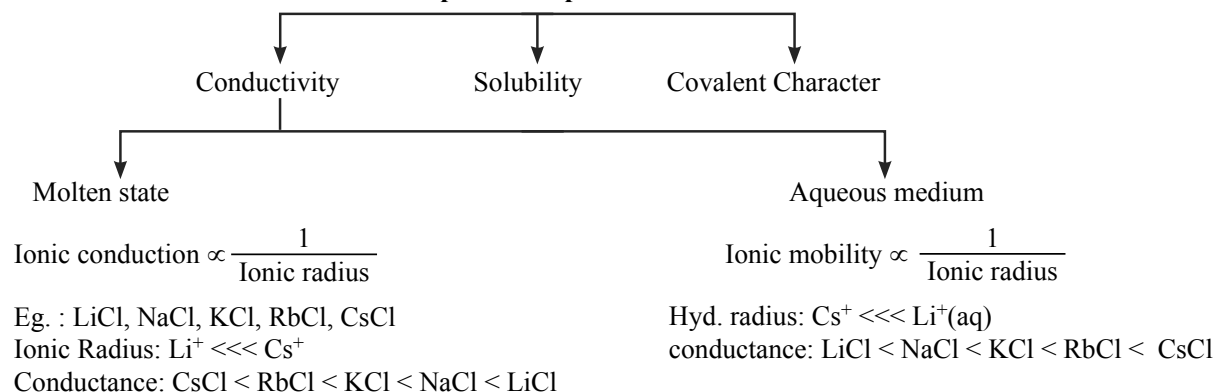
Hard, Solid, Brittle (strong electrostatic attraction between ions)

Non directional nature (ionic compound do not exhibit stereo isomerism)

Higher Melting Point / Boiling Point

Isomorphism:  $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ ;  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ ;  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

#### Specific Properties



### Hybridisation

**Table :** Hybridisation and geometry

| S.No. | Type or orbital | No. of hybrid orbital | 3D orientation | Example                          |
|-------|-----------------|-----------------------|----------------|----------------------------------|
| 1.    | one s + one p   | 2; sp                 | Linear         | $\text{BeH}_2$ , $\text{BeCl}_2$ |
| 2.    | one s + two p   | 3; sp <sup>2</sup>    | Triangular     | $\text{BCl}_3$ , $\text{BF}_3$   |

|    |                           |                                   |                        |                                    |
|----|---------------------------|-----------------------------------|------------------------|------------------------------------|
| 3. | one s + three p           | 4; sp <sup>3</sup>                | Tetrahedral            | CH <sub>4</sub> , CCl <sub>4</sub> |
| 4. | one s + three p + one d   | 5; sp <sup>3</sup> d              | Triangular bipyramidal | PCl <sub>5</sub>                   |
| 5. | one s + three p + two d   | 6; sp <sup>3</sup> d <sup>2</sup> | Octahedral             | SF <sub>6</sub>                    |
| 6. | one s + three p + three d | 7; sp <sup>3</sup> d <sup>3</sup> | Pentagonal bipyramidal | IF <sub>7</sub>                    |

## Type of Hybridisation & Possible Structure

Table : Hybridisation and Shape

| Type of Hybridisation                                | No. of B.P. | No. of L.P. | Shape                                                       | Examples                                                                                                                                                                                                                          |
|------------------------------------------------------|-------------|-------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. sp-hybridisation                                  | 2           | -           | Linear                                                      | BeF <sub>2</sub> , CO <sub>2</sub> , CS <sub>2</sub> , BeCl <sub>2</sub>                                                                                                                                                          |
| 2. (a) sp <sup>2</sup> -hybridisation                | 3           | -           | Trigonal planar                                             | BF <sub>3</sub> , AlCl <sub>3</sub> , BeF <sub>3</sub> <sup>-</sup>                                                                                                                                                               |
| (b) sp <sup>2</sup> -hybridisation                   | 2           | 1           | V-shape, Angular                                            | NO <sub>2</sub> <sup>-</sup> , SO <sub>2</sub> , O <sub>3</sub>                                                                                                                                                                   |
| 3. (a) sp <sup>3</sup> -hybridisation                | 4           | 0           | Tetrahedral                                                 | CH <sub>4</sub> , CCl <sub>4</sub> , PCl <sub>4</sub> <sup>+</sup> , ClO <sub>4</sub> <sup>-</sup> , NH <sub>4</sub> <sup>+</sup> , BF <sub>4</sub> <sup>-</sup> , SO <sub>4</sub> <sup>2-</sup> , AlCl <sub>4</sub> <sup>-</sup> |
| (b) sp <sup>3</sup> -hybridisation                   | 3           | 1           | Pyramidal                                                   | NH <sub>3</sub> , PF <sub>3</sub> , ClO <sub>3</sub> <sup>-</sup> , H <sub>3</sub> O <sup>+</sup> , PCl <sub>3</sub> , XeO <sub>3</sub> , N(CH <sub>3</sub> ) <sub>3</sub> , CH <sub>3</sub> <sup>-</sup>                         |
| (c) sp <sup>3</sup> -hybridisation                   | 2           | 2           | V-shape<br>Angular                                          | H <sub>2</sub> O, H <sub>2</sub> S, NH <sub>2</sub> <sup>-</sup> , OF <sub>2</sub> , Cl <sub>2</sub> O <sub>2</sub> , SF <sub>2</sub> , I <sub>3</sub> <sup>+</sup>                                                               |
| 4. (a) sp <sup>3</sup> d-hybridisation               | 5           | -           | Trigonal bipyramidal                                        | PCl <sub>5</sub> , SOF <sub>4</sub> , AsF <sub>5</sub>                                                                                                                                                                            |
| (b) sp <sup>3</sup> d-hybridisation                  | 4           | 1           | See-Saw, folded square distorted tetrahedral                | SF <sub>4</sub> , PF <sub>4</sub> <sup>-</sup> , AsF <sub>4</sub> <sup>-</sup> , SbF <sub>4</sub> <sup>-</sup> , XeO <sub>2</sub> F <sub>2</sub>                                                                                  |
| (c) sp <sup>3</sup> d-hybridisation                  | 3           | 2           | Almost T-shape                                              | ClF <sub>3</sub> , ICl <sub>3</sub>                                                                                                                                                                                               |
| (d) sp <sup>3</sup> d-hybridisation                  | 2           | 3           | Linear                                                      | I <sub>3</sub> <sup>-</sup> , Br <sub>3</sub> <sup>-</sup> , ICl <sub>2</sub> <sup>-</sup> , XeF <sub>2</sub>                                                                                                                     |
| 5. (a) sp <sup>3</sup> d <sup>2</sup> -hybridisation | 6           | -           | Square bipyramidal/octahedral                               | PCl <sub>6</sub> <sup>-</sup> , SF <sub>6</sub>                                                                                                                                                                                   |
| (b) sp <sup>3</sup> d <sup>2</sup> -hybridisation    | 5           | 1           | Square pyramidal/distorted octahedral                       | XeOF <sub>4</sub> , ClF <sub>5</sub> , SF <sub>5</sub> <sup>-</sup>                                                                                                                                                               |
| (c) sp <sup>3</sup> d <sup>2</sup> -hybridisation    | 4           | 2           | Square planar                                               | XeF <sub>5</sub> <sup>+</sup> , XeF <sub>4</sub>                                                                                                                                                                                  |
| 6. (a) sp <sup>3</sup> d <sup>3</sup> -hybridisation | 7           | -           | Pentagonal bipyramidal                                      | IF <sub>7</sub>                                                                                                                                                                                                                   |
| (b) sp <sup>3</sup> d <sup>3</sup> -hybridisation    | 6           | 1           | Pentagonal pyramidal/distorted octahedral/capped octahedral | XeF <sub>6</sub>                                                                                                                                                                                                                  |
| (c) sp <sup>3</sup> d <sup>3</sup> -hybridisation    | 5           | 2           | Pentagonal planar                                           | XeF <sub>5</sub> <sup>-</sup>                                                                                                                                                                                                     |

## Molecular Orbital Theory

Formation of molecule or orbitals from atomic orbitals ( $\psi_A \pm \psi_B$ )

$$\psi_{MO} = \psi_A \pm \psi_B$$

$$\sigma = \psi_A + \psi_B \text{ (Bonding molecular orbital)}$$

$$\sigma^* = \psi_A - \psi_B \text{ (Antibonding molecular orbital)}$$

**Relative energies** of M.O having

Less than or equal to 14 electrons.

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < [\pi^* 2p_x = \pi^* 2p_y] < \sigma^* 2p_z$$

For more than 14 electrons

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < [\pi 2p_x = \pi 2p_y] < [\pi^* 2p_x = \pi^* 2p_y] < \sigma^* 2p_z$$

**Bond order**

$$\text{Bond order} = \frac{1}{2} [\text{Number of bonding electrons} - \text{Number of antibonding electrons}]$$

$$\text{or, } = \frac{N_b - N_a}{2}$$

**Table:** Hybridisation of following species in specified state

| Species                          | Cationic part                                | Anionic part                                 |
|----------------------------------|----------------------------------------------|----------------------------------------------|
| $\text{PCl}_5$                   | $\text{PCl}_4^+$ ( $\text{sp}^3$ )           | $\text{PCl}_6^-$ ( $\text{sp}^3\text{d}^2$ ) |
| $\text{PBr}_5$                   | $\text{PBr}_4^+$ ( $\text{sp}^3$ )           | $\text{Br}^-$                                |
| $\text{XeF}_6$                   | $\text{XeF}_5^+$ ( $\text{sp}^3\text{d}^2$ ) | $\text{F}^-$                                 |
| $\text{N}_2\text{O}_5$           | $\text{NO}_2^+$ ( $\text{sp}$ )              | $\text{NO}_3^-$ ( $\text{sp}^2$ )            |
| $\text{I}_2\text{Cl}_6$ (liquid) | $\text{ICl}_2^+$ ( $\text{sp}^3$ )           | $\text{ICl}_4^-$ ( $\text{sp}^3\text{d}^2$ ) |
| $\text{Cl}_2\text{O}_6$          | $\text{ClO}_2^+$ ( $\text{sp}^2$ )           | $\text{ClO}_4^-$ ( $\text{sp}^3$ )           |
| $\text{I}_2$ (molten state)      | $\text{I}_3^+$ ( $\text{sp}^3$ )             | $\text{I}_3^-$ ( $\text{sp}^3\text{d}$ )     |

## Types of Intermolecular Forces

### Types of Intermolecular Forces

- Ion - Dipole
- Ion - Induced dipole
- Dipole - dipole
- Hydrogen bonding
- Dipole - Induced dipole
- Dispersion (London force)

## Force

**Table:** Relative strength of forces

| Force                            | Relative strength                | Interaction energy                                                                                                                         |
|----------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Ion-ion (Cation-anion)           | Very strong                      | Interaction energy $\propto r^{-1}$                                                                                                        |
| Covalent bonds                   | Strong, (140–523 KJ/mole)        | Interaction energy $\propto r^{-1}$                                                                                                        |
| Ion-dipole                       | Moderate                         | Interaction energy $\propto r^{-2}$                                                                                                        |
| Hydrogen bond                    | Moderate to weak (10–100 kJ/mol) | Interaction energy $\propto r^{-1}$                                                                                                        |
| Dipole-dipole                    | Weak (5–25 kJ/mol)               | Interaction energy $\propto r^{-3}$ (for stationary polar molecules)<br>Interaction energy $\propto r^{-6}$ (for rotating polar molecules) |
| London force or dispersion force | Very weak (0.05–20 kJ/mol)       | Interaction energy $\propto r^{-6}$                                                                                                        |

**Table:** Magnetic behaviour of odd electron species

|                                                          | Hybridisation | Shape           | Magnetic behaviour |
|----------------------------------------------------------|---------------|-----------------|--------------------|
| $\text{NO}_2$                                            | $\text{sp}^2$ | V shape         | Para               |
| $\text{ClO}_2$                                           | $\text{sp}^2$ | V shape         | Para               |
| $\text{ClO}_3$                                           | $\text{sp}^3$ | Pyramidal       | Para               |
| $\cdot\text{CH}_3$                                       | $\text{sp}^2$ | Trigonal planar | Para               |
| $\cdot\text{CH}_3/\cdot\text{CHF}_2/\text{CH}_2\text{F}$ | $\text{sp}^3$ | Pyramidal       | Para               |

**Odd  $e^-$  species:** Total number of electron or valence electron in odd number.